QUESTION 1 NAIVE BAYES

1 Classes

C_1 = (GoingOut = Yes) C_2 = (GoingOut = No)

2 Prior probabilities

15 records, with 8 in C_1 and 7 in C_2.

$$P(C_1) = \frac{8}{15} \qquad P(C_2) = \frac{7}{15}$$

3 Features

$X = (x_1, x_2, x_3, x_4)$

x_1 = SoreThroat (Symptoms)

x_2 = Normal (BodyTemp)

x_3 = False (Vaccinated)

x_4 = Partial (LockdownStatus)

4 Conditional probabilities

$P(x_i \mid C_j)$ = matching records in C_j ÷ total records in C_j. See Table 1 for the conditional probabilities.

Table 1. Conditional probabilities for Naive Bayes

Observed feature	$P(x_i \mid C_1)$	$P(x_i \mid C_2)$
x_1	3/8	2/7
x_2	3/8	1/7
x_3	1/8	4/7
x_4	1/8	2/7

5 Prediction

$$P(C_j) \times P(X \mid C_j) = P(C_j) \times \prod_{i=1}^{4} P(x_i \mid C_j)$$

$$P(C_1) \times P(X \mid C_1) = \frac{8}{15} \times \frac{3}{8} \times \frac{3}{8} \times \frac{1}{8} \times \frac{1}{8} = 0.001171875$$

$$P(C_2) \times P(X \mid C_2) = \frac{7}{15} \times \frac{2}{7} \times \frac{1}{7} \times \frac{4}{7} \times \frac{2}{7} \approx 0.003109815$$

$$P(X) = \sum_{j=1}^{2} P(C_j) \times P(X \mid C_j) \approx 0.001171875 + 0.003109815 = 0.004281690$$

$$P(C_1 \mid X) = \frac{P(C_1) \times P(X \mid C_1)}{P(X)} \approx \frac{0.001171875}{0.004281690} \approx 27.37\%$$

$$P(C_2 \mid X) = \frac{P(C_2) \times P(X \mid C_2)}{P(X)} \approx \frac{0.003109815}{0.004281690} \approx 72.63\%$$

Since 72.63% > 27.37%, the predicted class is **C_2 (GoingOut = No).**

QUESTION 1 GAUSSIAN NAIVE BAYES

1 Features

One-hot encode the four features of the sample to be predicted, then standardize the encoded values using the training means and standard deviations. The transformed sample is $Z = (z_1, \ldots, z_{11})$.

2 Conditional densities

For encoded feature i in class C_j, use its standardized value z_i and the class mean μ_{ij} and variance σ^2_{ij}. See Table 2 for the conditional densities.

$$f(z_i \mid C_j) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} \exp\left(-\frac{(z_i - \mu_{ij})^2}{2\sigma_{ij}^2}\right)$$

Table 2. Conditional densities for Gaussian Naive Bayes

Standardized feature	Encoded value	Z-score	$f(z_i \mid C_1)$	$f(z_i \mid C_2)$
z_1 = Symptoms_Cough	0	-0.81650	0.29907	0.27143
z_2 = Symptoms_Fever	0	-0.60302	0.34488	0.31973
z_3 = Symptoms_SoreThroat	1	1.41421	0.16882	0.11927
z_4 = BodyTemp_High	0	-0.93541	0.38907	0.12622
z_5 = BodyTemp_Mild	0	-0.60302	0.26996	0.46385
z_6 = BodyTemp_Normal	1	1.65831	0.15837	0.02510
z_7 = Vaccinated_False	1	1.41421	0.01717	0.26119
z_8 = Vaccinated_True	0	-1.41421	0.01717	0.26119
z_9 = LockdownStatus_Free	0	-1.22474	0.10071	0.27143
z_{10} = LockdownStatus_Full	0	-0.50000	0.44925	0.28921
z_{11} = LockdownStatus_Partial	1	2.00000	0.01457	0.10120

3 Prediction

Multiply the 11 densities in each class column from top to bottom. The products use unrounded values.

$$f(Z \mid C_1) \approx 0.29907 \times 0.34488 \times 0.16882 \times 0.38907 \times 0.26996 \times 0.15837 \times 0.01717 \times 0.01717 \times 0.10071 \times 0.44925 \times 0.01457 \approx 5.63054 \times 10^{-11}$$

$$f(Z \mid C_2) \approx 0.27143 \times 0.31973 \times 0.11927 \times 0.12622 \times 0.46385 \times 0.02510 \times 0.26119 \times 0.26119 \times 0.27143 \times 0.28921 \times 0.10120 \approx 8.24437 \times 10^{-9}$$

$$P(C_1) \times f(Z \mid C_1) \approx \frac{8}{15} \times (5.63054 \times 10^{-11}) \approx 3.00296 \times 10^{-11}$$

$$P(C_2) \times f(Z \mid C_2) \approx \frac{7}{15} \times (8.24437 \times 10^{-9}) \approx 3.84737 \times 10^{-9}$$

$$f(Z) \approx 3.00296 \times 10^{-11} + 3.84737 \times 10^{-9} \approx 3.87740 \times 10^{-9}$$

$$P(C_1 \mid Z) \approx \frac{3.00296 \times 10^{-11}}{3.87740 \times 10^{-9}} \approx 0.77\% \qquad P(C_2 \mid Z) \approx \frac{3.84737 \times 10^{-9}}{3.87740 \times 10^{-9}} \approx 99.23\%$$

Since 99.23% > 0.77%, the predicted class is **C_2 (GoingOut = No).**

QUESTION 1 DISCUSSION

1 Prediction results

Both methods used the same 15 training records. Table 3 compares the two predictions.

Table 3. Comparison of Naive Bayes predictions

Method	Yes	No	Prediction
Hand calculation	27.37%	72.63%	No
GaussianNB	0.77%	99.23%	No

2 Why the probabilities differ

Both methods use the same class priors. The hand calculation estimates likelihoods from the frequencies of the four observed category values within each class. GaussianNB fits a separate Gaussian density to each of the 11 columns produced by one-hot encoding and Z-score standardization.

For a continuous feature, a Gaussian curve can estimate a density at a value that did not appear in the training data. This works best when the feature is approximately normally distributed within each class. In this assignment, the original features are categorical. The two vaccination columns encode the same status, so treating them as independent can give vaccination more weight and help explain the higher posterior probability for No.

QUESTION 2 DECISION TREE

1 Data and training

A DecisionTreeClassifier predicted customer churn (Yes or No) from 19 features, including demographics, tenure, services, contract type, billing, and charges. Each categorical feature was encoded in one integer column. The 11 blank TotalCharges values were filled with zero. A stratified 80/20 split produced 5,634 training records and 1,409 test records.

The script reads 1,078 parameter combinations from a separate CSV and evaluates them in parallel using five-fold stratified cross-validation. Fixed random seeds make the results reproducible. The best combination used the Gini criterion, a maximum depth of 10, at least 100 samples to split a node, and at least 30 samples per leaf. Its mean validation accuracy was 79.41%. The script then retrained this model on the full training set.

2 Evaluation

Table 4. Confusion matrix on the test set

Actual class	Predicted Yes	Predicted No	Class accuracy
Yes	186	188	49.73%
No	111	924	89.28%

From Table 4, **test accuracy = (186 + 924) ÷ 1,409 = 78.78%.** This was 5.32 percentage points above the 73.46% baseline of predicting No for every customer. The model still missed 188 of the 374 customers who churned.

3 Discussion

The final tree had 80 leaves. Its first split used Contract at a threshold of 0.50, separating month-to-month contracts (code 0) from one- and two-year contracts (codes 1 and 2). Table 5 shows the training-set churn rates for both branches.

Table 5. Churn rates by root-node contract group

Contract group	Training records	Churn rate
Month-to-month	3,102	42.75%
One or two years	2,532	6.67%

The 36.08 percentage-point gap shows a strong association between contract type and churn in the training data. Contract had the highest feature importance (0.4639), followed by online security (0.1227) and tenure (0.0940). Table 6 lists the split-based importance of all 19 features.

Table 6. Decision tree feature importance

Feature	Score	Feature	Score	Feature	Score
Contract	0.4639	PaymentMethod	0.0156	SeniorCitizen	0.0001
OnlineSecurity	0.1227	MultipleLines	0.0139	OnlineBackup	0.0001
tenure	0.0940	StreamingMovies	0.0056	Partner	0.0000
TotalCharges	0.0897	TechSupport	0.0051	PhoneService	0.0000
MonthlyCharges	0.0825	DeviceProtection	0.0041	gender	0.0000
InternetService	0.0805	StreamingTV	0.0023		
PaperlessBilling	0.0177	Dependents	0.0020		